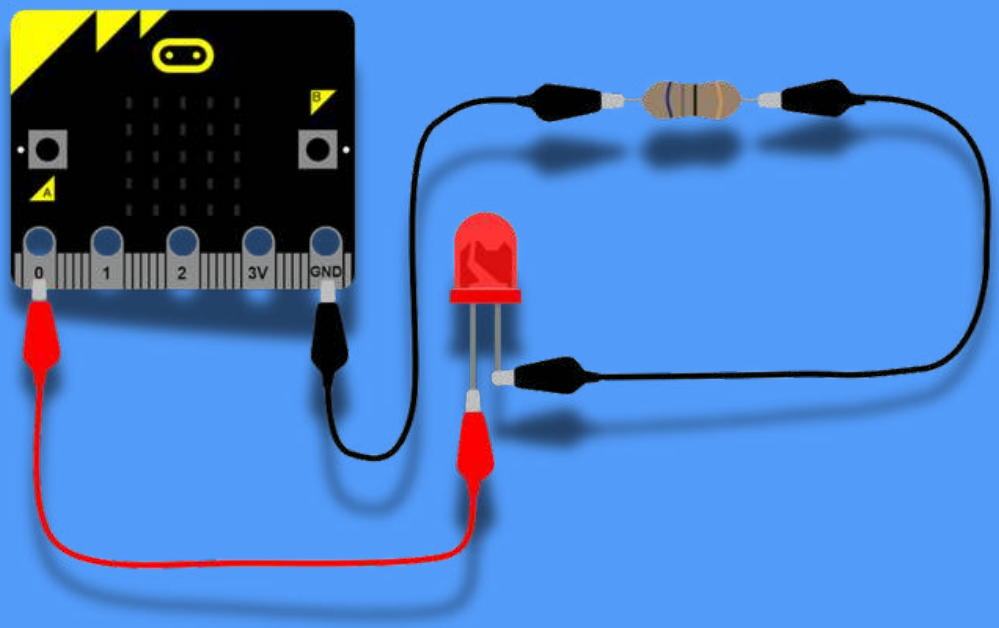
Lesson 1

micro:bit Course

Module: LED & Motor Minigolf Challenge

Connect Micro:bit with a LED and Program it



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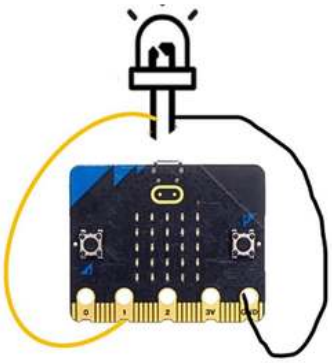
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Connect Micro:bit with a LED and Program it

How to connect:

What you need:

- 1 USB cable
- 1 Micro:bit
- 1 LED
- 2 crocodile cables



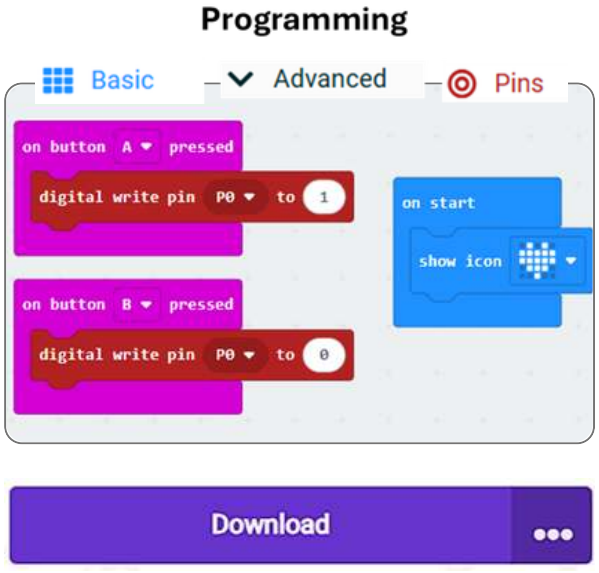
Further info

See how to connect your LED in this video (Part 1)

Now connect the LED to the micro.bit

- **Long leg** of the LED -> digital output (**P1**)
- **Short leg** of the LED -> ground (**GND**)

How to program:



Further info

See how to program your LED in this video (Part 2)

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Connect Micro:bit with a LED and Program it



Program a continuous flashing light

- Create a program where the light flashes using "**digital write pin**", "**forever**" and "**pause**" blocks.

Varying Flash Speed

- Create a program where the flash speed of the LED can be adjusted using the Micro:bit's **accelerometer** (e.g., tilting the Micro:bit changes the delay).

Light Sensor Integration

- Use a **light sensor**, and have the LED turn on only when the light level falls below a certain threshold.

Random Flashing

- Use the random number block to make the LED flash at random intervals.

Debugging

- Intentionally **introduce errors** (e.g., reversed LED polarity, incorrect pin connections) and try to troubleshoot and fix them.



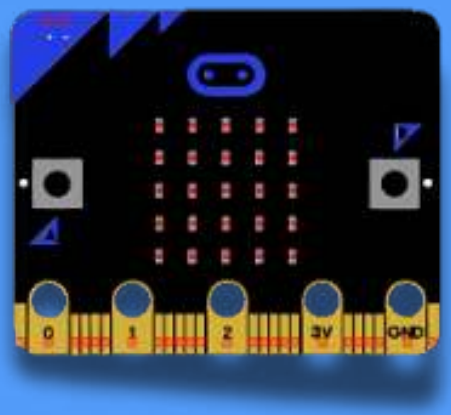
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Lesson 2

micro:bit Course

Module: LED & Motor Minigolf Challenge

Connect Micro:bit with the Expansion Board





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Connect Micro:bit with the Expansion Board

The **micro:bit** is a powerful mini-computer, but to truly bring your creations to life, you'll need to connect external components like motors, lights and sensors. That's where the **Elecfreaks ring:bit v2** expansion board come in. This lesson will guide you through connecting the ring:bit v2 to your micro:bit, unlocking a world of possibilities without the hassle of unreliable crocodile clips.

Need to Know!

- **More connections:** Your micro:bit has some "**ports**" (holes) where you can connect things. The expansion board gives you even more of these ports, so you can connect more lights, motors, or sensors.
- **Easy connections:** Instead of having to connect tiny wires, the board changes some of the micro:bit's ports (P0, P1, P2) into "**GVS ports.**" Think of GVS ports as easier and simpler plugs that many devices use.
- **Power boost:** The **Ring:bit** can hold three AAA batteries. This lets you power things like a little robot car or other electronic parts without needing to plug into a wall. So it makes your micro:bit mobile.



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Connect Micro:bit with the Expansion Board

How to connect?

To connect the expansion board, carefully align the micro:bit so that the pins match. Then, use screws to fasten the micro:bit to the board through the provided mounting holes.

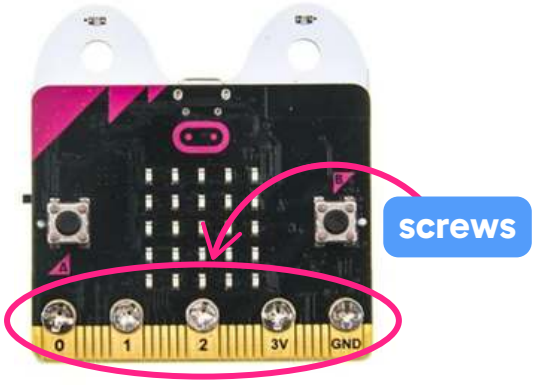
Further info

See how to connect the expansion board



Expansion Board





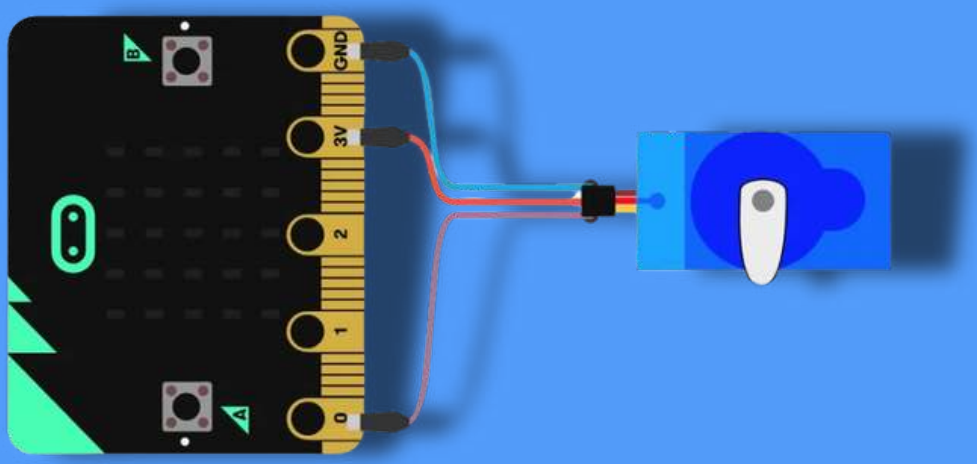
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Lesson 3

micro:bit Course

Module: LED & Motor Minigolf Challenge

How to Connect and Program a 180 Servo Motor



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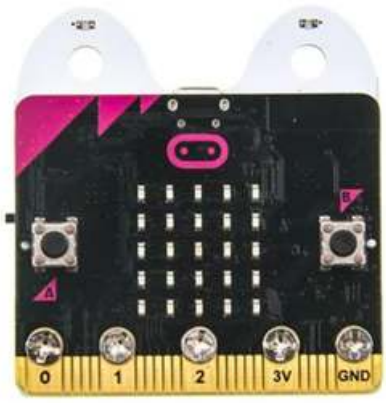
How to Connect and Program a 180 Servo Motor

What do you need?

180° Servo Motor



Micro:bit connected with expansion board

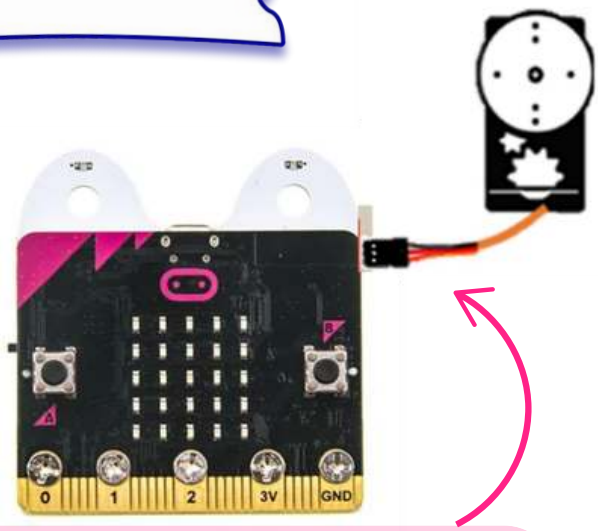


How to connect?

Further info

See how to connect your micro:bit in this video



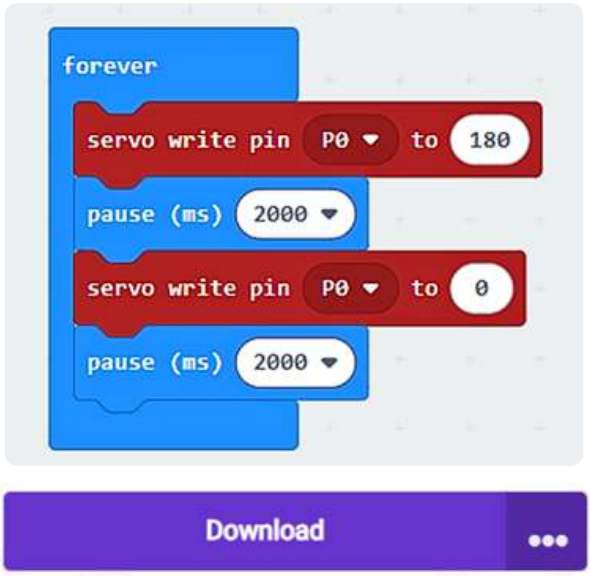


Connecting your servo motor is simple: plug it into the GV0 port on the expansion board.

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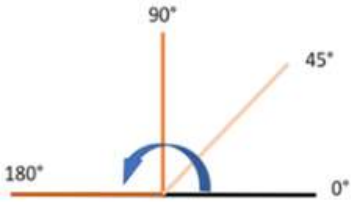
How to Connect and Program a 180 Servo Motor

How to program?



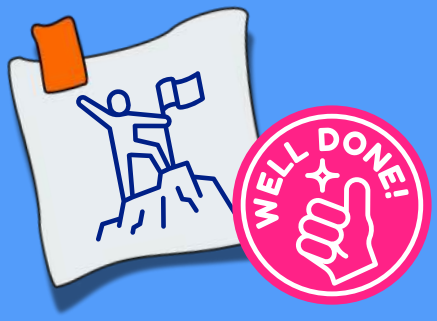
Further info

See how to program your micro:bit in this video



CHALLENGE

Try to use the A and B buttons to control your motor.



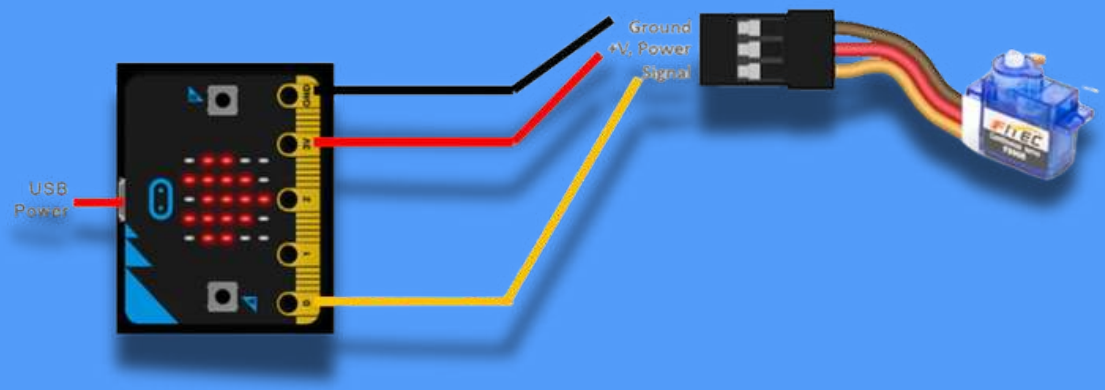
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Lesson 4

micro:bit Course

Module: LED & Motor Minigolf Challenge

How to Connect and Program a 360 Servo Motor



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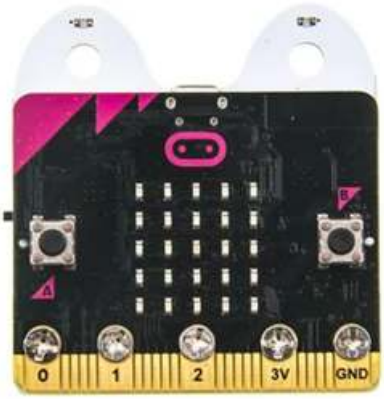
How to Connect and Program a 360 Servo Motor

What do you need?

360° Servo Motor



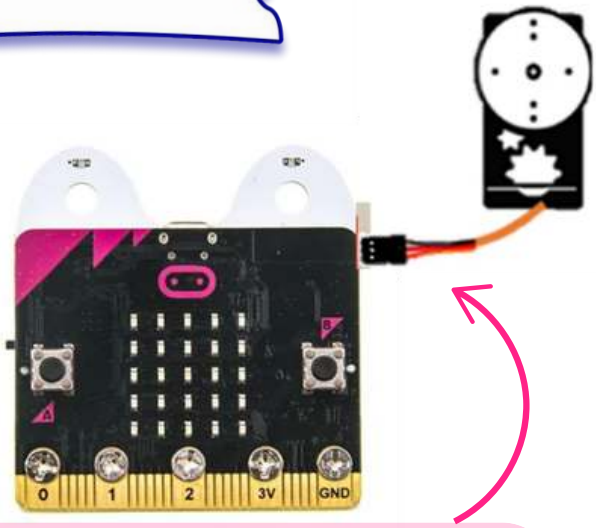
Micro:bit connected with expansion board



How to connect?

Further info

See how to connect your micro:bit in this video

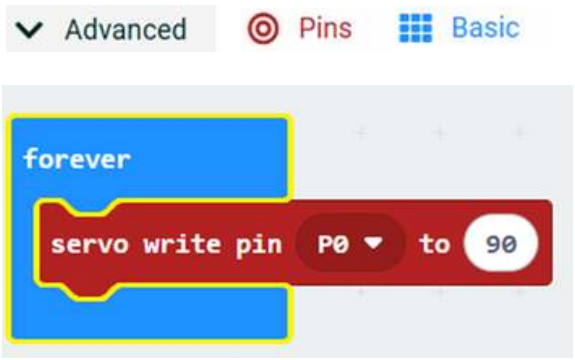


Connecting your servo motor is simple: plug it into the GV0 port on the expansion board.

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How to Connect and Program a 360 Servo Motor

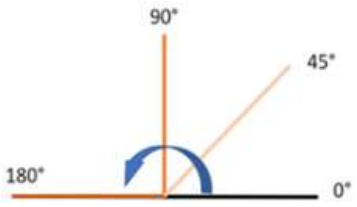
How to program?



0 = fullspeed

90 = stop

180 = fullspeed





- Change the number to 0 and then to 180 to see what happens
- Pick different numbers between 0 and 180 and see what happens



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micro:bit Course

Module: LED & Motor Minigolf Challenge

Lesson 5

RGB Lights



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RGB Lights

What do you need?

1 micro: Circular RGB LED Expansion Board

1 USB-cable

1 Micro:bit



Further info

- See how to connect your RGB Lights
- See the video for coding instructions
- Click here to see the code

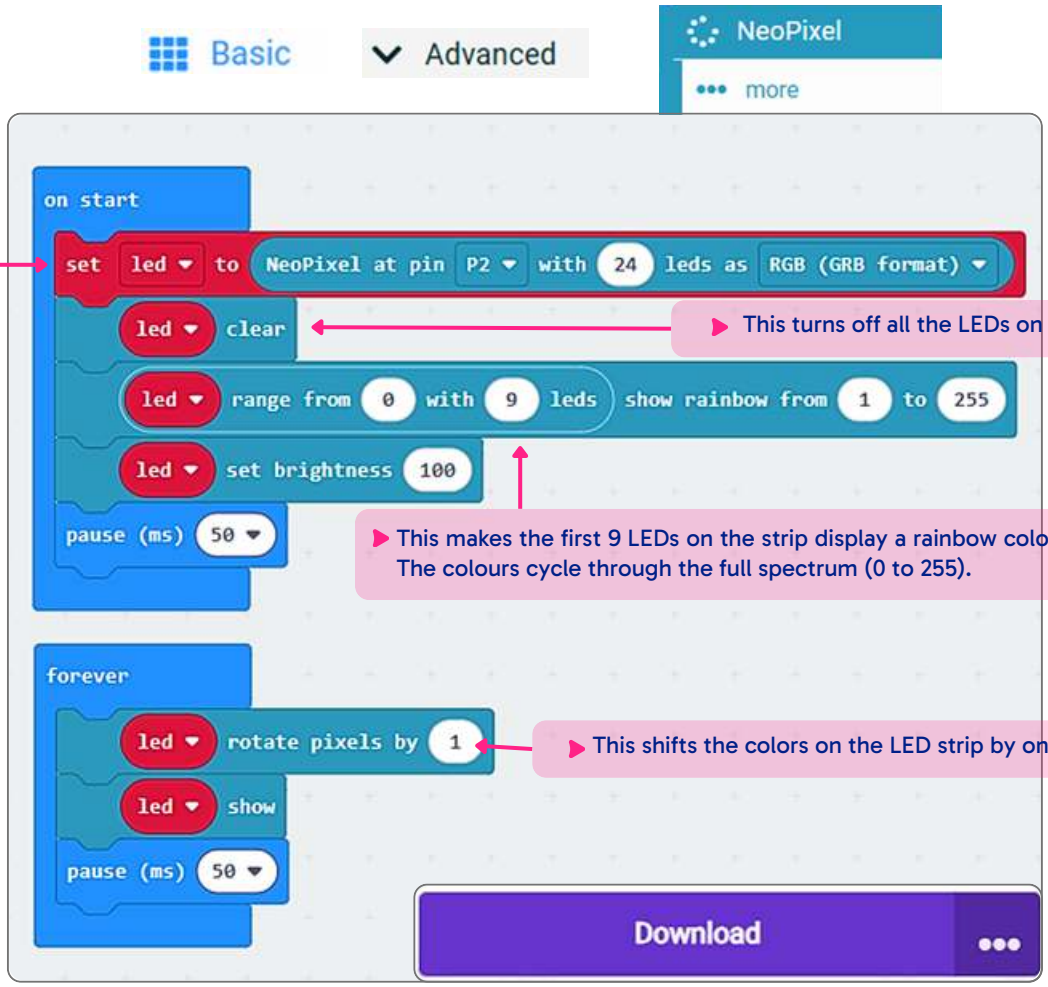
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RGB Lights

How to program?

Level 1: Effects

First, nine lights show nine different colour changes. Then, those same nine colours move around a ring of lights, like a flowing waterfall of colours, over and over.



This turns off all the LEDs on the strip.

This makes the first 9 LEDs on the strip display a rainbow colour pattern. The colours cycle through the full spectrum (0 to 255).

This shifts the colors on the LED strip by one position.

- This line initializes the NeoPixel LED strip. It tells the micro:bit: Which pin the LED strip is connected to (P2).
- How many LEDs are in the strip (24).
- That they are RGB LEDs (capable of displaying various colors).
- The color order (GRB - Green, Red, Blue).

Attention!

1. **To download the program, you must connect the Micro:bit to your computer.**
2. **To illuminate the board, you must connect it to the battery pack.**
3. **Now you can disconnect the Micro:bit from the computer.**

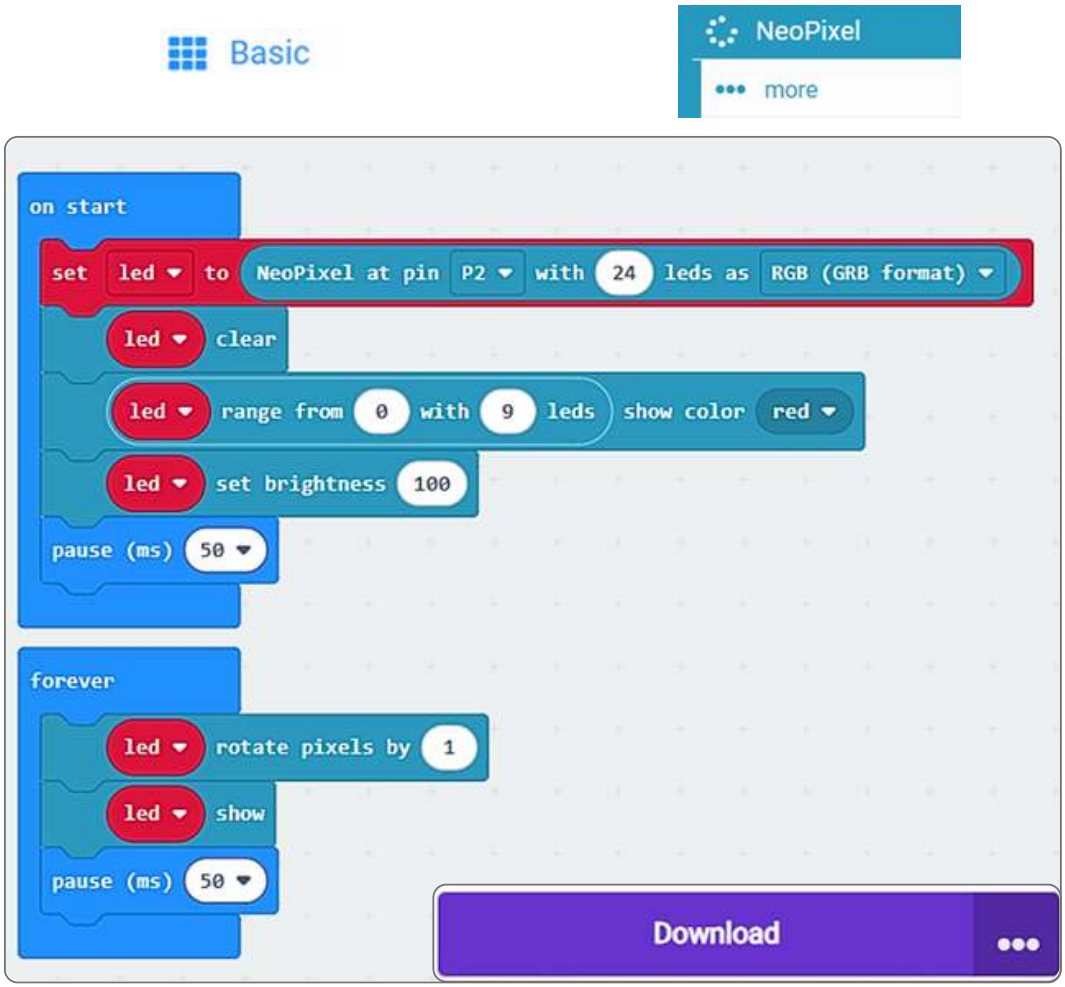
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RGB Lights

How to program?

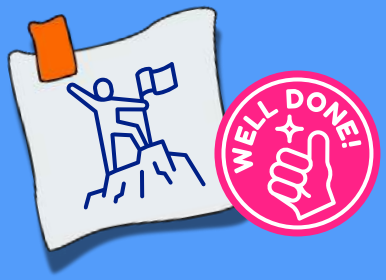
Level 2: Your Colour

Make the lights turn on in order, from the very first one to the last one in the colour you like!



CHALLENGE

Change the colour in your code and see what happens!



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Lesson 6

micro:bit Course

Module: LED & Motor Minigolf Challenge

Challenge: Design a Mini Golf



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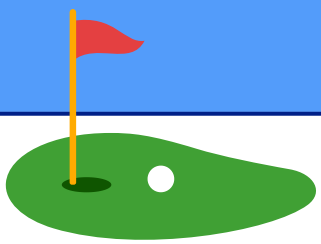
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Challenge: Design a Mini Golf



Alright everyone, get ready to put your creativity to the test!

You will be using recycled materials to build a mini-golf course that includes moving obstacles, cool lights, fun sound effects, and a clear start and end.

Let's see how you can combine your knowledge and imagination to create an awesome mini-golf experience!

What do you need?

- Micro:bit
- 180° and 360° servo motors
- LED and RGB lights
- Recycling materials
- Ping-Pong balls



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Challenge: Design a Mini Golf



Sketch your Minigolf here:

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